

Lesson 5: Vector dot and cross products

vector dot product and vector length

Addition

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_n + b_n \end{bmatrix}$$

Scalar multiplication

$$c \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} ca_1 \\ ca_2 \\ \vdots \\ ca_n \end{bmatrix}$$

Dot product

$\vec{a} \cdot \vec{b}$

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \cdot \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \overbrace{a_1 b_1 + a_2 b_2 + a_3 b_3 + \dots + a_n b_n}^{\text{scalar}}$$

$$\begin{bmatrix} 2 \\ 5 \end{bmatrix} \cdot \begin{bmatrix} 7 \\ 1 \end{bmatrix} = 2 \cdot 7 + 5 \cdot 1 = 14 + 5 = 19$$

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ 0 \\ 5 \end{bmatrix}$$

$$1 \cdot (-2) + 2 \cdot 0 + 3 \cdot 5 = -2 + 0 + 15 = 13$$

"length"

$$\|\vec{a}\| = \sqrt{a_1^2 + a_2^2 + \dots + a_n^2}$$

$$\vec{a} \cdot \vec{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \cdot \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = a_1^2 + a_2^2 + \dots + a_n^2$$

$$\|\vec{a}\| = \sqrt{\vec{a} \cdot \vec{a}}$$

$$\boxed{\|\vec{a}\|^2 = \vec{a} \cdot \vec{a}}$$

proving vector dot product properties

$$\vec{v} \cdot \vec{w} = \vec{w} \cdot \vec{v} \quad \checkmark$$

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \quad \vec{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} \quad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$\vec{v} \cdot \vec{w} = v_1 w_1 + v_2 w_2 + \dots + v_n w_n$$

$$\vec{w} \cdot \vec{v} = w_1 v_1 + w_2 v_2 + \dots + w_n v_n$$

$$v_1 w_1 = w_1 v_1$$

commutative property.

$$(\vec{v} + \vec{w}) \cdot \vec{x} = \vec{v} \cdot \vec{x} + \vec{w} \cdot \vec{x} \quad \checkmark$$

$$\vec{v} + \vec{w} = \begin{bmatrix} v_1 + w_1 \\ v_2 + w_2 \\ \vdots \\ v_n + w_n \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$(v_1 + w_1)x_1 + (v_2 + w_2)x_2 + \dots + (v_n + w_n)x_n$$

$$= (v_1 x_1 + w_1 x_1) + (v_2 x_2 + w_2 x_2) + \dots$$

$$\vec{v} \cdot \vec{x} = v_1 x_1 + v_2 x_2 + \dots + v_n x_n$$

$$\vec{w} \cdot \vec{x} = w_1 x_1 + w_2 x_2 + \dots + w_n x_n$$

$$\vec{v} \cdot \vec{x} + \vec{w} \cdot \vec{x} = (v_1 x_1 + w_1 x_1) + (v_2 x_2 + w_2 x_2) + \dots + (v_n x_n + w_n x_n)$$

$$(c\vec{v}) \cdot \vec{w} \stackrel{!}{=} c(\vec{v} \cdot \vec{w}) \quad \checkmark$$

$$\begin{bmatrix} cv_1 \\ cv_2 \\ \vdots \\ cv_n \end{bmatrix} \cdot \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} = cv_1 w_1 + cv_2 w_2 + \dots + cv_n w_n$$

$$c(\vec{v} \cdot \vec{w}) = c(v_1 w_1 + v_2 w_2 + \dots + v_n w_n) \\ = cv_1 w_1 + cv_2 w_2 + \dots + cv_n w_n$$

proof of the Cauchy-Schwarz inequality.

$$\left. \begin{array}{l} \vec{x}, \vec{y} \in \mathbb{R}^n \quad |\vec{x} \cdot \vec{y}| \leq \|\vec{x}\| \|\vec{y}\| \\ \text{non zero} \quad |\vec{x} \cdot \vec{y}| = \|\vec{x}\| \|\vec{y}\| \Leftrightarrow \vec{x} = c\vec{y} \end{array} \right\}$$

Cauchy-Schwarz inequality

$$p(t) = \|\vec{y} - t\vec{x}\|^2 \geq 0$$

$$= (\vec{y} - t\vec{x}) \cdot (\vec{y} - t\vec{x})$$

$$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$$

$$= \vec{y} \cdot \vec{y} - t\vec{x} \cdot \vec{y} - t\vec{y} \cdot \vec{x} + t^2 \vec{x} \cdot \vec{x}$$

$$= (\vec{y} \cdot \vec{y}) - 2(\vec{x} \cdot \vec{y})t + t^2 (\vec{x} \cdot \vec{x}) \geq 0$$

$$p(t) = at^2 - bt + c \geq 0$$

$$b^2/4a - \frac{2b^2}{4a} + c \geq 0$$

$$p(b/2a) = a \frac{b^2}{4a} - \frac{b \cdot b}{2a} + c \geq 0$$

$$-\frac{b^2}{4a} + c \geq 0 \quad c \geq b^2/4a \rightarrow 4ac \geq b^2$$

$$4(\|\vec{y}\|^2 \|\vec{x}\|^2) \geq (2(\vec{x} \cdot \vec{y}))^2$$

$$4\|\vec{y}\|^2 \|\vec{x}\|^2 \geq 4(\vec{x} \cdot \vec{y})^2$$

$$\|\vec{y}\| \|\vec{x}\| \geq |\vec{x} \cdot \vec{y}| \quad \leftarrow \text{Cauchy-Schwarz inequality}$$

$$\vec{x} = c\vec{y}$$

$$|\vec{x} \cdot \vec{y}| = |c\vec{y} \cdot \vec{y}| = |c| |\vec{y} \cdot \vec{y}| = |c| \|\vec{y}\|^2$$

$$= |c| \|\vec{y}\| \|\vec{y}\|$$

$$= \|\vec{x}\| \|\vec{y}\|$$

$$= \|\vec{x}\| \|\vec{y}\|$$

vector triangle inequality.

Cauchy-Schwarz Inequality

$$\vec{x}, \vec{y} \in \mathbb{R}^n$$

non-zero $|\vec{x} \cdot \vec{y}| \leq \|\vec{x}\| \|\vec{y}\|$

$$\vec{x} = c\vec{y} \quad \vec{x} \cdot \vec{y} = \|\vec{x}\| \|\vec{y}\|$$

↑
non zero

$$\vec{x} \cdot \vec{y} \leq |\vec{x} \cdot \vec{y}|$$

$$\|\vec{x} + \vec{y}\|^2 = (\vec{x} + \vec{y}) \cdot (\vec{x} + \vec{y})$$

$$= \vec{x} \cdot (\vec{x} + \vec{y}) + \vec{y} \cdot (\vec{x} + \vec{y})$$

$$= \vec{x} \cdot \vec{x} + \vec{x} \cdot \vec{y} + \vec{y} \cdot \vec{x} + \vec{y} \cdot \vec{y}$$

$$= \|\vec{x}\|^2 + 2(\vec{x} \cdot \vec{y}) + \|\vec{y}\|^2$$

$$\|\vec{x} + \vec{y}\|^2 \leq \|\vec{x}\|^2 + 2\|\vec{x}\| \|\vec{y}\| + \|\vec{y}\|^2$$

$$\|\vec{x} + \vec{y}\|^2 \leq (\|\vec{x}\| + \|\vec{y}\|)^2$$

$$\|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\| \quad \text{Triangle inequality.}$$

$$\|\vec{x} + \vec{y}\| \geq \|\vec{x}\| + \|\vec{y}\|$$

↑
 $\vec{x} = c\vec{y} \quad c > 0$

positive

$$\vec{x} = c\vec{y}$$

$$\vec{x} \cdot \vec{y} = (\vec{x} \cdot \vec{y})$$

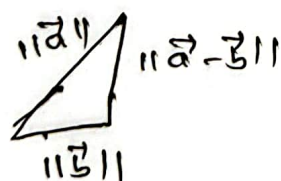
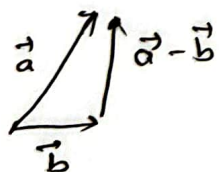
$$= c\vec{y} \cdot \vec{y}$$

$$= c\|\vec{y}\|^2$$

Defining the angles between vectors

$\|\vec{a}\|$ = length $\Rightarrow$ scalar

"angle" between vectors



Reasons why I couldn't

$$\|\vec{b}\| > \|\vec{a}\| + \|\vec{a} - \vec{b}\|$$

$$\|\vec{a}\| > \|\vec{a} - \vec{b}\| + \|\vec{b}\|$$

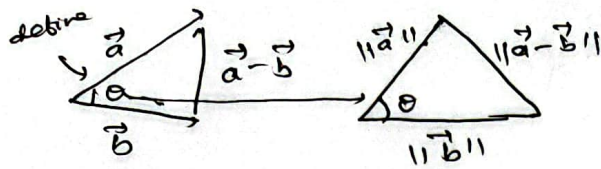
$$\|\vec{a} - \vec{b}\| > \|\vec{a}\| + \|\vec{b}\|$$

$$\|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\|$$

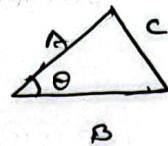
$$\|\vec{a}\| = \|\vec{b} + (\vec{a} - \vec{b})\| \leq \|\vec{b}\| + \|\vec{a} - \vec{b}\|$$

$$\|\vec{b}\| = \|\vec{a} + (\vec{b} - \vec{a})\| \leq \|\vec{a}\| + \|\vec{b} - \vec{a}\|$$

$$\|\vec{a} - \vec{b}\| = \|\vec{a} + (-\vec{b})\| \leq \|\vec{a}\| + \|\vec{b}\|$$



Law of cosine



$$c^2 = A^2 + B^2 - 2AB \cos \theta$$

$$\|\vec{a} - \vec{b}\|^2 = \|\vec{b}\|^2 + \|\vec{a}\|^2 - 2\|\vec{a}\|\|\vec{b}\|\cos \theta$$

$$(\vec{a} - \vec{b}) \cdot (\vec{a} - \vec{b}) =$$

$$\vec{a} \cdot \vec{a} - \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b}$$

$$\|\vec{a}\|^2 - 2(\vec{a} \cdot \vec{b}) + \|\vec{b}\|^2 = \|\vec{b}\|^2 + \|\vec{a}\|^2 - 2\|\vec{a}\|\|\vec{b}\|\cos \theta$$

$$\boxed{(\vec{a} \cdot \vec{b}) = \|\vec{a}\|\|\vec{b}\|\cos \theta} \leftarrow \text{angle "between" them}$$

$$\vec{a} = c\vec{b} \quad c > 0 \Rightarrow \theta = 0 \quad (\text{collinear})$$

$$c < 0 \Rightarrow \theta = 180^\circ$$

perpendicular: θ between $\vec{a}, \vec{b}$ is 90°
 $\vec{a} \cdot \vec{b}$

$$\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\|\cos 90^\circ$$

$$\vec{a} \cdot \vec{b} = 0$$

$$\vec{a}, \vec{b} \text{ perpendicular} \Rightarrow \vec{a} \cdot \vec{b} = 0$$

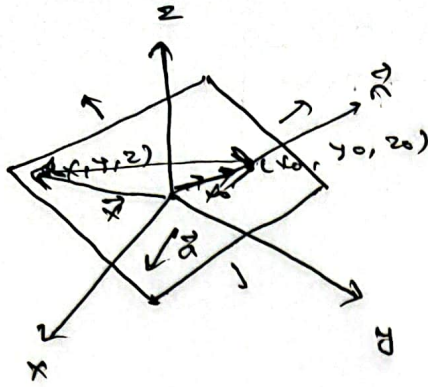
'0 cannot be defined for 0 vector'

$$\vec{a} \cdot \vec{b} = 0 \Rightarrow \text{"orthogonal"}$$

zero vector is orthogonal to everything

Defining a plane in $\mathbb{R}^3$ with a point and normal vector

Equation of a plane in $\mathbb{R}^3$



$\vec{n}$ = "normal"

perpendicular to everything in the plane

$\vec{a}$ = lies on the plane

$\vec{b}$ = " " " "

$$\vec{n} \cdot \vec{a} = 0$$

$\vec{n} = (x_0, y_0, z_0)$ definition

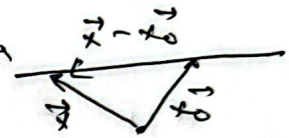
$\Downarrow$

$$Ax + By + Cz = 0$$

$$\vec{x}_0 = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$(\vec{x} - \vec{x}_0)$ $\swarrow$ perpendicular to $\vec{n}$



$$\vec{n} = \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix}$$

$$\vec{n} \cdot (\vec{x} - \vec{x}_0) = 0$$

$$\begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix} \cdot \begin{bmatrix} x - x_0 \\ y - y_0 \\ z - z_0 \end{bmatrix} = 0$$

$$Ax + By + Cz = D$$

$$n_1(x - x_0) + n_2(y - y_0) + n_3(z - z_0) = 0$$

$$\vec{n} = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix} \quad \vec{x}_0 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad \vec{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\vec{x} - \vec{x}_0 = \begin{bmatrix} x-1 \\ y-2 \\ z-3 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix} \cdot \begin{bmatrix} x-1 \\ y-2 \\ z-3 \end{bmatrix} = 0$$

$$x - 1 + 3y - 6 - 2z + 6 = 0$$

$$x + 3y - 2z = 1$$

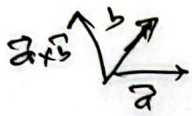
Cross product introduction

Dot product $\vec{a}, \vec{b} \in \mathbb{R}^n \Rightarrow \text{scalar}$

cross product only defined in $\mathbb{R}^3 \Rightarrow \text{vector}$

$$\vec{a} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \quad \vec{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \quad \vec{a} \times \vec{b} = \begin{bmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ -7 \\ 1 \end{bmatrix} \times \begin{bmatrix} 5 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} -7 \cdot 4 - 1 \cdot 2 \\ 1 \cdot 5 - 1 \cdot 4 \\ 1 \cdot 2 - (-8) \cdot 1 \end{bmatrix} = \begin{bmatrix} -30 \\ 1 \\ 10 \end{bmatrix}$$



$\vec{a}, \vec{b}$
orthogonal $\Rightarrow \vec{a} \cdot \vec{b} = 0$

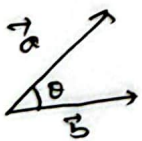
$$\begin{bmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{bmatrix} \cdot \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

$$= a_1 a_2 b_3 - a_1 a_3 b_2 + a_2 a_3 b_1 - a_2 a_1 b_3 + a_3 a_1 b_2 - a_3 a_2 b_1$$

$$= 0$$

Dot and cross product comparison / intuition

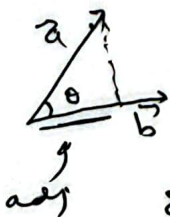
$$\vec{a} \cdot \vec{b} = \|\vec{a}\| \|\vec{b}\| \cos \theta$$



$$\frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \|\vec{b}\|} = \cos \theta$$

$$\theta = \arccos \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \|\vec{b}\|}$$

$$\boxed{\|\vec{b}\| \|\vec{a}\| \cos \theta}$$



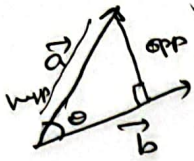
$$\vec{a} \cdot \vec{b} = \|\vec{b}\| \cdot \text{adj}$$

$$\cos \theta = \frac{\text{adj}}{\|\vec{a}\|} \Rightarrow \|\vec{a}\| \cos \theta = \text{adj}$$

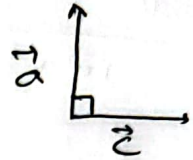
$$\vec{a} \cdot \vec{b} = \|\vec{a}\| \|\vec{b}\| \cos \theta$$

dot $\Rightarrow$ product of "lengths" of vectors
"same direction"

$$\|\vec{a} \times \vec{b}\| = \|\vec{a}\| \|\vec{b}\| \sin \theta = \|\vec{b}\| \underbrace{\|\vec{a}\| \sin \theta}_{\text{opp}}$$

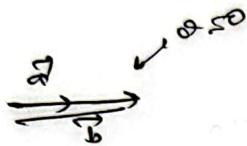


$$\sin \theta = \frac{\text{opp}}{\text{hyp}} = \frac{\text{opp}}{\|\vec{a}\|}$$



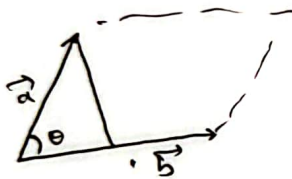
$$\|\vec{a}\| \sin \theta = \text{opp}$$

$$\|\vec{a} \times \vec{b}\| = \underbrace{\|\vec{a}\| \|\vec{b}\|}_{\vec{a} \cdot \vec{b} \cos \theta}$$



$$\|\vec{a} \times \vec{b}\| = 0$$

$$\vec{a} \cdot \vec{b} = \|\vec{a}\| \|\vec{b}\|$$

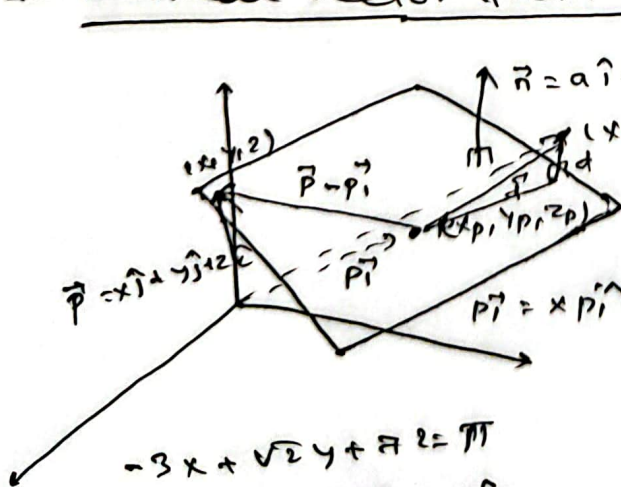


$$\text{Area} = \|\vec{b}\| \cdot \text{height}$$

$$\text{Area} = \|\vec{b}\| \|\vec{a}\| \sin \theta$$

$$\sin \theta = \frac{\text{height}}{\|\vec{a}\|} \Rightarrow \text{height} = \|\vec{a}\| \sin \theta$$

Normal vector from plane equation



$$-3x + \sqrt{2}y + z = 7$$

$$\vec{n} = -3\hat{i} + \sqrt{2}\hat{j} + \hat{k}$$

$$ax_p + by_p + cz_p = a + b + c = 7$$

$$ax + by + cz = a + b + c = 7$$

$$Ax + By + Cz = D$$

$$\vec{n} = A\hat{i} + B\hat{j} + C\hat{k}$$

$$-D = -Ax_p - By_p - Cz_p$$

point distance to plane

$$\vec{r} = (x_0 - x_p)\hat{i} + (y_0 - y_p)\hat{j} + (z_0 - z_p)\hat{k}$$

$$\cos \theta = \frac{d}{|\vec{r}|}$$

$$|\vec{r}| \cos \theta = d$$

$$\frac{|\vec{n} \cdot \vec{r}|}{|\vec{n}|} = d$$

$$\frac{\vec{n} \cdot \vec{r}}{|\vec{n}|} = d$$

$$\vec{n} \cdot \vec{r} = \frac{Ax_0 - Ax_p + By_0 - By_p + Cz_0 - Cp}{\sqrt{A^2 + B^2 + C^2}}$$

$$\frac{Ax_0 + By_0 + Cz_0 - D}{\sqrt{A^2 + B^2 + C^2}}$$